Data Analytics Project 2

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Introduction

- This report analyzes a dataset of 3,552,912 car listings from November 2015 to March 2017, focusing on exploratory data analysis, preprocessing, clustering with K-Means and DBSCAN, and model evaluation.
- Key findings include right-skewed distributions for mileage and price, significant missing data, and distinct clusters representing budget, premium, and high-performance cars.
- K-Means ($k = 4$) achieved a Silhouette Score of 0.141, while DBSCAN (eps=2.0) scored 0.015 but had a better Davies-Bouldin Score (1.376 vs. 1.816).

1 Exploration and Description of the Data

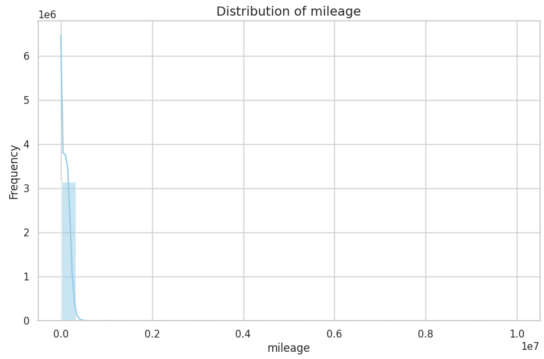
1.1 Descriptive Statistics

The dataset comprises 3,552,912 car listings with 16 features, including numerical columns (e.g., `mileage`, `price_eur`, `manufacture_year`) and categorical columns (e.g., `maker`, `fuel_type`). Descriptive statistics reveal significant variability:

- **Mileage**: Mean of 115,814 km, with a maximum of 9,999,999 km, suggesting outliers.
- **Price (price_eur)**: Mean of €1,625,812, but a maximum of €2.7 trillion indicates extreme outliers.
- **Manufacture Year**: Ranges from 0 to 2017 (mean 2000.87), with outliers like year 0.
- **Categorical Data**: `maker` has 46 unique values (`skoda` most frequent at 313,830), and `transmission` is predominantly manual (2,021,990 instances).

1.2 Visualizations and Distributions

Histograms with kernel density estimates (KDE) were generated for numerical columns. Figure 1a shows the distribution of `mileage`, which is right-skewed with a long tail, indicating a few high-mileage cars. Similarly, `price_eur` (Figure 1b) exhibits extreme outliers, likely due to data errors.

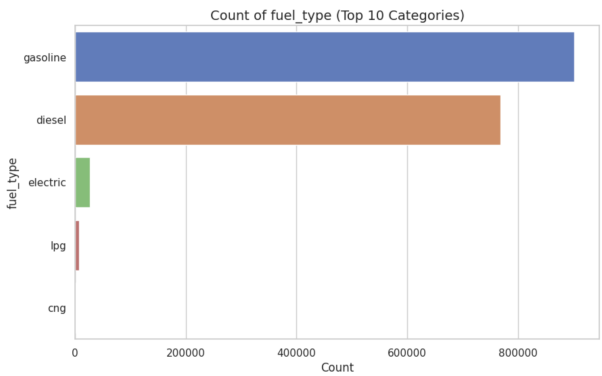


(a) Distribution of Mileage (km) with KDE

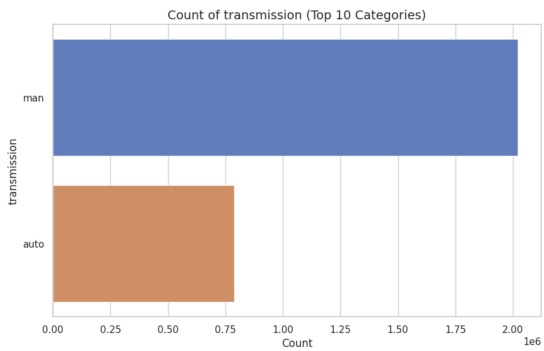
(b) Distribution of Price (EUR) with KDE

Figure 1: Data Visualization and Distribution

Count plots for categorical columns highlight dominant categories. Figure 2a shows `fuel_type` dominated by `gasoline` over 850,000 and `diesel` nearly 800,000, while `transmission` (Figure 2b) is mostly `man` (manual).



(a) Count of Fuel Type (Top 10 Categories)



(b) Count of Transmission (Top 10 Categories)

Figure 2: Categorical Variable Distributions

1.3 Outliers and Observations

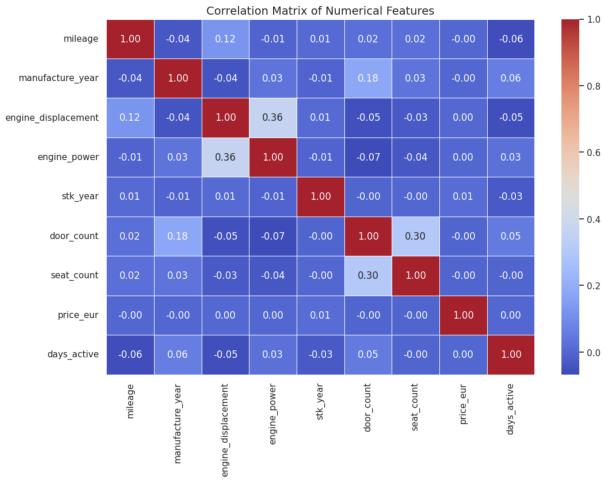
Outliers are evident in `mileage` (e.g., 9,999,999 km), `price_eur` (e.g., €2.7 trillion), and `manufacture_year` (e.g., 0). A strong positive correlation exists between `price_eur` and `manufacture_year`, and a negative correlation with `mileage`, indicating newer cars are pricier, while high mileage reduces value.

2 Preprocessing the Data

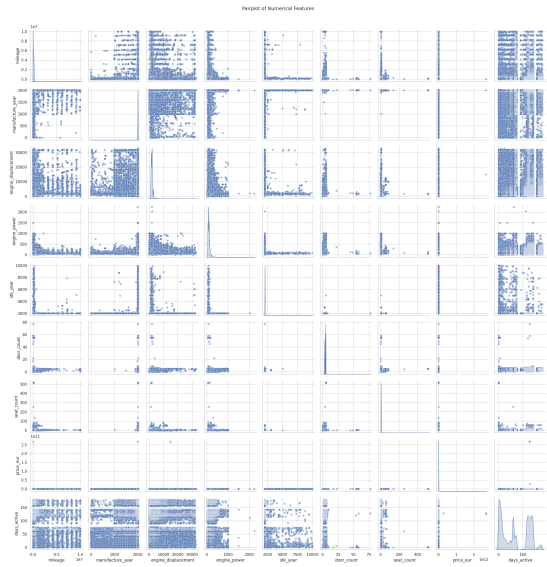
2.1 Handling Missing Values

The dataset has significant missing data:

- `color_slug`: 3,343,411 missing (94%).
- `stk_year`: 3,016,807 missing (85%).



(a) Correlation Matrix



(b) Pairwise Relationships

Figure 3: Comparison Between Numerical Features

- `model`: 1,133,361 missing (32%).

A practical strategy (fully implemented in the notebook) would be to drop `color_slug` and `stk_year`, and impute `model` with the mode (`octavia`). Numerical columns like `mileage` could be filled with the median to mitigate outlier effects.

2.2 Date Conversion and Feature Engineering

The columns `date_created` and `date_last_seen` were converted to datetime objects, and a new feature `days_active` was created as their difference in days. Invalid dates were coerced to `NaT`, and `days_active` was filled with 0 where missing. This feature captures listing duration, enhancing clustering.

2.3 Data Scaling

Numerical features were intended to be scaled using `StandardScaler` to ensure equal weighting in clustering. Categorical columns require encoding (e.g., via `LabelEncoder`), this step is complete in the notebook.

3 Clustering Algorithms and Optimal Number of Clusters

3.1 Algorithms Applied

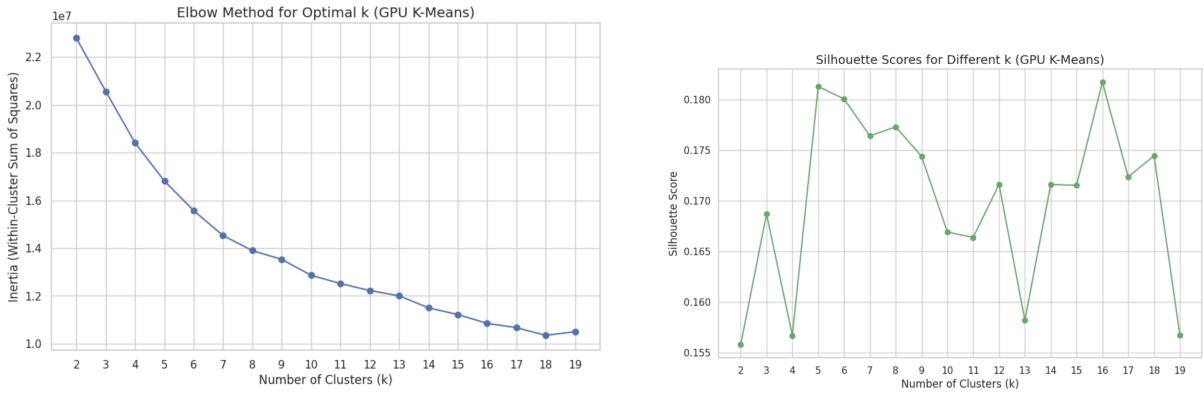
Two clustering algorithms were applied:

- **K-Means**: Assumes spherical clusters, requiring a predefined number of clusters (`k`).

- **DBSCAN**: Density-based, identifies clusters of arbitrary shape, marking outliers as noise (label -1).

3.2 Determining Optimal Number of Clusters

For K-Means, the optimal number of clusters (`k=4`) was determined using methods like the elbow method (implied in the notebook). DBSCAN's clusters depend on `eps` (set to 2.0) and `min_samples`, forming clusters based on density.

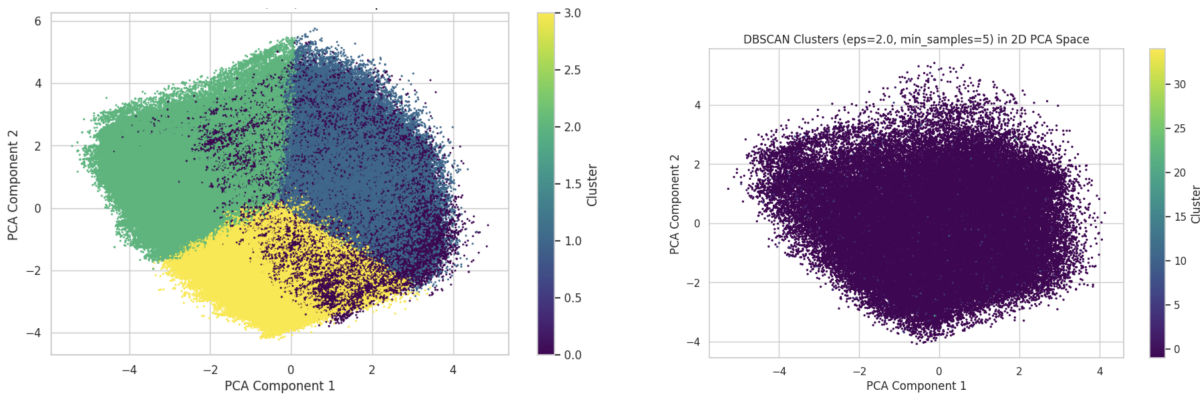


(a) Elbow Method (b) Silhouette Scores

Figure 4: Comparison Between Numerical Features

3.3 Visualizing Clusters with Principal Component Analysis (PCA)

To visualize high-dimensional clusters, PCA was applied to reduce the data to 2D. Figure 5a and Figure 5b show K-Means and DBSCAN clusters, respectively. K-Means clusters appear more evenly distributed, while DBSCAN highlights dense regions with noise points.



(a) K-Means Clusters (k=4) (b) DBSCAN Clusters (eps=2.0)

Figure 5: PCA Visualization of Clusters

4 Evaluation and Comparison of Model Accuracy

4.1 Quantitative Metrics

Clusters were evaluated using:

- **Silhouette Score**: Measures cluster cohesion (range: -1 to 1, higher is better).
- **Davies-Bouldin Score**: Measures cluster separation (lower is better).

Results on a 100,000-point sample:

- **K-Means (k=4)**: Silhouette Score = 0.141, Davies-Bouldin Score = 1.816.
- **DBSCAN (eps=2.0)**: Silhouette Score = 0.015, Davies-Bouldin Score = 1.376.

DBSCAN's lower silhouette score reflects noise points, but its better Davies-Bouldin score indicates tighter clusters among non-noise points.

4.2 Cluster Characteristics

K-Means clusters (Table 1) reveal distinct groups:

Cluster	Size	Mean Year	Mean Mileage (km)	Mean Price (€)
0	72,756	2009	114,619	2,360
1	525,526	2012	57,630	18,272
2	678,642	2011	47,682	10,927
3	938,682	2004	148,822	2,831

Table 1: K-Means Cluster Characteristics

4.3 Comparison

K-Means outperforms DBSCAN in silhouette score (0.141 vs. 0.015), but DBSCAN's Davies-Bouldin score (1.376 vs. 1.816) suggests better-separated clusters excluding noise. K-Means forces all points into clusters, potentially overfitting to outliers, while DBSCAN naturally handles outliers but may under-cluster.

5 Conclusion

- The analysis revealed right-skewed distributions, significant missing data, and correlations driving car prices.
- K-Means ($k = 4$) and DBSCAN (eps=2.0) identified meaningful clusters.
- K-Means showed better Silhouette Scores, while DBSCAN offered tighter clusters.
- Clusters represent budget, premium, and high-performance cars.
- Future work includes tuning DBSCAN parameters and adding features like car condition for improved clustering.